# MA 17: HOW TO SOLVE IT
# PROBLEM SET 4

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**Due:** 11:59PM of November 15. Late submissions will not be accepted.

***Instructions:***

- Please submit solutions to at most 3 problems, written carefully. You are encouraged to work on more, but the TA will provide feedback on only 3. I recommend selecting the 3 for which you most want feedback.
- Collaboration on homework before you have solved at least 3 problems independently is discouraged, though not strictly forbidden. You should make a serious effort to work through each problem independently before turning to classmates, using the internet, or asking AI.
- As long as your submission clearly shows attempts on at least 3 problems, you will earn 8 points toward the final grade. The correctness of your solutions will not affect your points, although feedback will be provided.
- Submission should be made through Canvas.

**Problem 1.**
For each integer $n \geq 0$, prove that the number of $3 \times 3$ matrices with nonnegative integer entries, all of whose row sums and column sums equal $n$, is

$$\frac{(n+1)(n+2)(n^2+3n+4)}{8}.$$

**Problem 2.** (Putnam 1967, A2)
Define $S_0$ to be 1. For $n \geq 1$, let $S_n$ be the number of $n \times n$ matrices whose elements are nonnegative integers with the property that $a_{ij} = a_{ji}$, $(i, j = 1, 2, \ldots, n)$ and where $\sum_{i=1}^{n} a_{ij} = 1$, $(j = 1, 2, \ldots, n)$. Prove

(a) $S_{n+1} = S_n + nS_{n-1}$,

(b) $\displaystyle\sum_{n=0}^{\infty} S_n \frac{x^n}{n!} = \exp(x + x^2/2), \quad \text{where } \exp(x) = e^x.$

**Problem 3.**
An alphabet has $n$ distinct letters, where $n$ is a positive integer. Prove that the number of words containing each letter exactly twice, with no two identical letters adjacent, is

$$\sum_{j=0}^{n} (-1)^j \binom{n}{j} \frac{(2n-j)!}{2^{n-j}}.$$

**Problem 4.** (Putnam 2023, B1)
Consider an $m$-by-$n$ grid of unit squares, indexed by $(i, j)$ with $1 \leq i \leq m$ and $1 \leq j \leq n$. There are $(m-1)(n-1)$ coins, which are initially placed in the squares $(i, j)$ with $1 \leq i \leq m-1$ and $1 \leq j \leq n-1$. If a coin occupies the square $(i, j)$ with $i \leq m-1$ and $j \leq n-1$ and the squares

$(i+1,j),(i,j+1)$, and $(i+1,j+1)$ are unoccupied, then a legal move is to slide the coin from $(i,j)$ to $(i+1,j+1)$. How many distinct configurations of coins can be reached starting from the initial configuration by a (possibly empty) sequence of legal moves?

**Problem 5.** (Putnam 2018, B6)
Let $S$ be the set of sequences of length 2018 whose terms are in the set $\{1,2,3,4,5,6,10\}$ and sum to 3860. Prove that the cardinality of $S$ is at most

$$2^{3860} \cdot \left(\frac{2018}{2048}\right)^{2018}.$$

**Problem 6.** (Putnam 2002, B1)
Shanille O'Keal shoots free throws on a basketball court. She hits the first and misses the second, and thereafter the probability that she hits the next shot is equal to the proportion of shots she has hit so far. What is the probability she hits exactly 50 of her first 100 shots?

**Problem 7.** (Putnam 1958 Spring, A3)
Real numbers are chosen at random from the interval $(0 \le x \le 1)$. If after choosing the $n$th number the sum of the numbers so chosen first exceeds 1, show that the expected or average value for $n$ is $e$.

**Problem 8.** (Putnam 2001, A2)
You have coins $C_1, C_2, \ldots, C_n$. For each $k$, $C_k$ is biased so that, when tossed, it has probability $1/(2k+1)$ of falling heads. If the $n$ coins are tossed, what is the probability that the number of heads is odd? Express the answer as a rational function of $n$.

**Problem 9.** (Putnam 2024, B4)
Let $n$ be a positive integer. Set $a_{n,0} = 1$. For $k \ge 0$, choose an integer $m_{n,k}$ uniformly at random from the set $\{1,\ldots,n\}$, and let

$$a_{n,k+1} = \begin{cases} a_{n,k}+1, & \text{if } m_{n,k} > a_{n,k}; \\ a_{n,k}, & \text{if } m_{n,k} = a_{n,k}; \\ a_{n,k}-1, & \text{if } m_{n,k} < a_{n,k}. \end{cases}$$

Let $E(n)$ be the expected value of $a_{n,n}$. Determine $\lim_{n\to\infty} E(n)/n$.

**Problem 10.** (Putnam 2020, A4)
Consider a horizontal strip of $N+2$ squares in which the first and the last square are black and the remaining $N$ squares are all white. Choose a white square uniformly at random, choose one of its two neighbors with equal probability, and color this neighboring square black if it is not already black. Repeat this process until all the remaining white squares have only black neighbors. Let $w(N)$ be the expected number of white squares remaining. Find

$$\lim_{N\to\infty} \frac{w(N)}{N}.$$

**Problem 11.** (Problem 8 in Handout 4)
Alice starts with $m$ coins and Bob with $n$ coins, where $m, n$ are positive integers. Each round, an independent fair coin toss decides who receives one coin from the other player. They stop when one player has no coins. Find the expected number of rounds.

**Problem 12.** (Putnam 2020, B3, Problem 9 in Handout 4)
Let $x_0 = 1$, and let $\delta$ be some constant satisfying $0 < \delta < 1$. Iteratively, for $n = 0, 1, 2, \ldots$, a point $x_{n+1}$ is chosen uniformly from the interval $[0, x_n]$. Let $Z$ be the smallest value of $n$ for which $x_n < \delta$. Find the expected value of $Z$, as a function of $\delta$.

**Reminder** Next Class: Nov 10.